Supplementary Information: Perovskite Solar Cell Performance Prediction with LLM-Derived Feature Augmentation under Leakage-Aware DOI Evaluation

Tier naming. Throughout the main paper and this document, t_1 = PDP-derived device, processing and composition features; $t_2 = t_1$ + curated material-property descriptors; $t_3 = t_2$ + the 36 LLM-derived recipe flags. An earlier, superseded 44-predictor literature representation (the "earlier literature block") is *not* t_3; where its results are shown they are labelled "t_2 + earlier lit. block". In the underlying run files and result tables of the repository this block carries the tier name `t3`; this document relabels it and never mixes its numbers with the recipe-flag results.

Table S1. Predictors per tier

Column sets shown for a preprocessor fitted on the full filtered data (documentation); in the experiments the `_na` indicators and the >80% missing filter are re-derived from each training partition. Tiers are nested: $t_1 \subset t_2 \subset t_3$ (77, 179 and 215 predictor columns in the common-split models).

t1: 77 predictors (77 added relative to the previous tier)

process (21): ETL_deposition_method, HTL_deposition_method, Perovskite_deposition_method, Perovskite_thickness_nm, ETL_thickness_nm, HTL_thickness_nm, Backcontact_thickness_nm, Perov_anneal_temp_C, Perov_anneal_time_min, Perov_deposition_ Perovskite_dimension_list_of_layers, Perov_quenching_crystallisation, Perovskite_dimension_3D_with_2D_capping_layer, Perovskite_deposition_solvent_annealing, Perovskite_thickness_nm_na, ETL_thickness_nm_na, HTL_thickness_nm_na, Backcontact_thickness_nm_na, Perov_anneal_temp_C_na, Perov_anneal_time_min_na, Perovskite_dimension_list_of_layers_na

device (9): Cell_architecture, Perovskite_band_gap_measured, is_flexible_x, is_semitransparent, Encapsulation, Add_lay_front, Perovskite_pl_max_nm, Perovskite_band_gap_measured_na, Perovskite_pl_max_nm_na

composition (47): num_a_ions, num_b_ions, num_c_ions, total_ion_count, a_coeff_sum, b_coeff_sum, c_coeff_sum, a_FA_fraction, a_MA_fraction, a_Cs_fraction, a_Rb_fraction, a_organic_fraction, a_other_fraction, b_Pb_fraction, b_Sn_fraction, b_Bi_fraction, b_other_fraction, halide_I_fraction, halide_Br_fraction, halide_Cl_fraction, halide_F_fraction halide_other_fraction, is_inorganic, is_leadfree, is_3D, is_2D, is_2D3D_mixture, Perovskite_composition_perovskite_inspired_ Perovskite_composition_perovskite_ABC3_structure, a_coeff_sum_na, b_coeff_sum_na, c_coeff_sum_na, a_FA_fraction_na, a_MA_fraction_na, a_Cs_fraction_na, a_Rb_fraction_na, a_organic_fraction_na, a_other_fraction_na, b_Pb_fraction_na, b_Sn_fraction_na, b_Bi_fraction_na, b_other_fraction_na, halide_I_fraction_na, halide_Br_fraction_na, halide_Cl_fraction_na halide_F_fraction_na, halide_other_fraction_na

t2: 179 predictors (102 added relative to the previous tier)

material (102): ETL_band_gap, ETL_LUMO, ETL_HOMO, ETL_electron_mobility, ETL_hole_mobility, ETL_dielectric_constant, ETL_min_atomic_mass, ETL_max_atomic_mass, ETL_avg_atomic_mass, ETL_min_radius, ETL_max_radius, ETL_avg_radius, ETL_en_min, ETL_en_max, ETL_en_avg, HTL_band_gap, HTL_LUMO, HTL_HOMO, HTL_electron_mobility, HTL_hole_mobility, HTL_dielectric_constant, HTL_min_atomic_mass, HTL_max_atomic_mass, HTL_avg_atomic_mass, HTL_min_radius, HTL_max_radius, HTL_avg_radius, HTL_en_min, HTL_en_max, HTL_en_avg, Perovskite_band_gap, Perovskite_LUMO, Perovskite_HOMO, Perovskite_electro Perovskite_hole_mobility, Perovskite_dielectric_constant, Perovskite_min_atomic_mass, Perovskite_max_atomic_mass, Perovskite_avg_atomic_mass, Perovskite_min_radius, Perovskite_max_radius, Perovskite_avg_radius, Perovskite_en_min, Perovskite_en_max, Perovskite_en_avg, perovskite_tolerance_factor, perovskite_octahedral_factor, perovskite_cation_radius_ perovskite_tau_descriptor, HTL_hole_barrier, ETL_electron_barrier, ETL_band_gap_na, ETL_LUMO_na, ETL_HOMO_na, ETL_electron_mobility_na, ETL_hole_mobility_na, ETL_dielectric_constant_na, ETL_min_atomic_mass_na, ETL_max_atomic_mass_na ETL_avg_atomic_mass_na, ETL_min_radius_na, ETL_max_radius_na, ETL_avg_radius_na, ETL_en_min_na, ETL_en_max_na, ETL_en_avg_na, HTL_band_gap_na, HTL_LUMO_na, HTL_HOMO_na, HTL_electron_mobility_na, HTL_hole_mobility_na, HTL_dielectric_con HTL_min_atomic_mass_na, HTL_max_atomic_mass_na, HTL_avg_atomic_mass_na, HTL_min_radius_na, HTL_max_radius_na, HTL_avg_radius_na, HTL_en_min_na, HTL_en_max_na, HTL_en_avg_na, Perovskite_band_gap_na, Perovskite_LUMO_na, Perovskite_HOMO_ Perovskite_electron_mobility_na, Perovskite_hole_mobility_na, Perovskite_dielectric_constant_na, Perovskite_min_atomic_ma

Perovskite_max_atomic_mass_na, Perovskite_avg_atomic_mass_na, Perovskite_min_radius_na, Perovskite_max_radius_na, Perovskite_avg_radius_na, Perovskite_en_min_na, Perovskite_en_max_na, Perovskite_en_avg_na, perovskite_tolerance_factor_na perovskite_octahedral_factor_na, perovskite_cation_radius_ratio_na, perovskite_tau_descriptor_na, HTL_hole_barrier_na, ETL_electron_barrier_na

t3: 215 predictors (36 recipe flags added to t_2)

Flags are paper-level, binary, and missing (native NaN, no indicator) for publications without an extraction record. Share positive among the 2,835 extracted papers in parentheses: rc_cl_source (20.5%), rc_acetate (4.7%), rc_scn (2.3%), rc_snf2 (0.7%), rc_ava (0.8%), rc_hx_acid (7.0%), rc_alkali (2.4%), rc_cs (12.0%), rc_pea_family (60.8%), rc_lewis_base (34.7%), rc_water_add (4.1%), rc_polymer (17.7%), rc_carbon (28.4%), rc_dio (0.7%), rc_htl_tbp (24.5%), rc_htl_litfsi (23.5%), rc_htl_fk (4.9%), rc_htl_f4tcnq (0.8%), rc_htl_cu (4.5%), rc_htl_borate (0.3%), rc_etl_ticl4 (5.6%), rc_etl_dopant (6.2%), rc_undoped (9.1%), rc_interface_pass (15.7%), rc_antisolvent (6.4%), rc_vacuum (17.6%), rc_two_step (3.9%), rc_hot_cast (0.0%), rc_solvent_anneal (1.0%), rc_flexible (4.3%), rc_tandem (3.4%), rc_lead_free (4.0%), rc_inorganic (7.5%), rc_mesoscopic (22.3%), rc_n2_box (4.0%), rc_ambient (10.4%).

Earlier literature block (44 predictors; not part of t_3)

literature (44): has_BCP, has_LiF, has_PEAI, has_SAM, has_C60, has_PCBM, has_Spiro, has_PEDOT, has_NiOx, has_SnO2, has_TiO2, has_ZnO, has_PTAA, has_fullerene, has_passivation_layer, is_flexible_y, is_encapsulated, has_stability_measurement, used_antisolvent, used_glovebox, used_vacuum_evaporation, json_numeric_values_found, has_BCP_na, has_LiF_na, has_PEAI_na, has_SAM_na, has_C60_na, has_PCBM_na, has_Spiro_na, has_PEDOT_na, has_NiOx_na, has_SnO2_na, has_TiO2_na, has_ZnO_na, has_PTAA_na, has_fullerene_na, has_passivation_layer_na, is_flexible_y_na, is_encapsulated_na, has_stability_measurement_na used_antisolvent_na, used_glovebox_na, used_vacuum_evaporation_na, json_numeric_values_found_na

Table S2. Per-split results of the matched CatBoost five-holdout experiment (DOI-grouped; t_3 in the run files = earlier literature block)

Subset	Tier	Split	n_{test}	R^2	MAE	RMSE
nip	t1	1	5476	0.3524	3.101	3.937
nip	t1	2	5376	0.3780	3.067	3.928
nip	t1	3	5688	0.3845	3.068	3.936
nip	t1	4	5785	0.3923	2.983	3.814
nip	t1	5	5736	0.3652	3.091	3.976
nip	t2	1	5476	0.4006	2.931	3.788
nip	t2	2	5376	0.4228	2.933	3.784
nip	t2	3	5688	0.4030	2.999	3.876
nip	t2	4	5785	0.4049	2.957	3.775
nip	t2	5	5736	0.3877	3.007	3.905
nip	t2 + earlier lit. block	1	5476	0.3939	2.960	3.809
nip	t2 + earlier lit. block	2	5376	0.4197	2.953	3.794
nip	t2 + earlier lit. block	3	5688	0.4091	2.982	3.856
nip	t2 + earlier lit. block	4	5785	0.4071	2.943	3.768
nip	t2 + earlier lit. block	5	5736	0.3963	2.981	3.877
overall	t1	1	8224	0.3707	3.082	3.914
overall	t1	2	8505	0.3345	3.101	3.958
overall	t1	3	8269	0.3604	3.136	3.945
overall	t1	4	8550	0.3990	3.036	3.859
overall	t1	5	8594	0.3767	3.051	3.862
overall	t2	1	8224	0.4145	2.943	3.775
overall	t2	2	8505	0.3712	2.985	3.847
overall	t2	3	8269	0.3787	3.067	3.888
overall	t2	4	8550	0.4420	2.892	3.719
overall	t2	5	8594	0.4326	2.864	3.684
overall	t2 + earlier lit. block	1	8224	0.4089	2.966	3.793
overall	t2 + earlier lit. block	2	8505	0.3731	2.982	3.842
overall	t2 + earlier lit. block	3	8269	0.3782	3.075	3.890
overall	t2 + earlier lit. block	4	8550	0.4408	2.893	3.723

Subset	Tier	Split	n_{test}	R^2	MAE	RMSE
overall	t2 + earlier lit. block	5	8594	0.4317	2.874	3.687
pin	t1	1	2437	0.3046	3.116	3.919
pin	t1	2	2521	0.1653	3.166	4.151
pin	t1	3	2723	0.2863	3.123	4.020
pin	t1	4	2872	0.2065	3.202	4.200
pin	t1	5	2441	0.2596	3.231	4.134
pin	t2	1	2437	0.3447	3.026	3.804
pin	t2	2	2521	0.2043	3.081	4.053
pin	t2	3	2723	0.3437	2.994	3.855
pin	t2	4	2872	0.2379	3.105	4.116
pin	t2	5	2441	0.3248	3.028	3.948
pin	t2 + earlier lit. block	1	2437	0.3592	2.980	3.762
pin	t2 + earlier lit. block	2	2521	0.1815	3.136	4.111
pin	t2 + earlier lit. block	3	2723	0.3385	3.009	3.871
pin	t2 + earlier lit. block	4	2872	0.2571	3.105	4.064
pin	t2 + earlier lit. block	5	2441	0.3315	3.026	3.928

Table S3. Split integrity (all 270 recorded fits)

DOI-grouped fits: 135; maximum DOI overlap between test and train: 0, validation and train: 0, test and validation: 0.

Records without DOI: 232 (one block). Publications (normalised DOIs): 7326.

Data SHA-256: `600efce587a1bd81c2a192e36db6534d8d78a3edfadb635dd40e685c9314ffdc`; config SHA-256: `dddb51a44e89fde8`; git HEAD at run time: `9b1d2ca3e1c7` (working tree dirty: True); CatBoost 1.2.10, scikit-learn 1.6.1, Python 3.13.5.

Table S4. Earlier literature block: coverage and constancy within publication

Feature	coverage	constant within DOI
contact_thickness_nm	0.176	1.000
etl_anneal_temp_C	0.053	1.000
etl_thickness_nm	0.101	1.000
has_BCP	0.362	1.000
has_C60	0.362	1.000
has_LiF	0.362	1.000
has_NiOx	0.362	1.000
has_PCBM	0.362	1.000
has_PEAI	0.362	1.000
has_PEDOT	0.362	1.000
has_PTAA	0.362	1.000
has_SAM	0.362	1.000
has_SnO2	0.362	1.000
has_Spiro	0.362	1.000
has_TiO2	0.362	1.000
has_ZnO	0.362	1.000
has_fullerene	0.362	1.000
has_passivation_layer	0.362	1.000
has_stability_measurement	0.362	1.000
htl_anneal_temp_C	0.037	1.000
htl_thickness_nm	0.081	1.000
is_encapsulated	0.362	1.000
is_flexible_y	0.362	1.000
json_numeric_values_found	0.362	1.000
perov_anneal_temp_C	0.070	1.000
perov_anneal_time_min	0.063	1.000
perov_concentration	0.051	1.000
perovskite_thickness_nm	0.122	1.000
spin_speed_rpm	0.100	1.000
used_antisolvent	0.362	1.000
used_glovebox	0.362	1.000
used_vacuum_evaporation	0.362	1.000

Coverage values below the median belong to sparsely reported process parameters; all other columns are non-missing exactly for the 14,896 records of publications with an extraction record.

Table S5. Earlier 44-predictor literature block: five-holdout ablation (matched CatBoost, DOI-grouped; mean ± s.d. over five splits)

This experiment evaluated the superseded literature block, not t_3. ΔR^2 is the mean per-split difference to t_2.

Subset	Tier	R^2	MAE	RMSE	ΔR^2 vs t_2
Pooled	t_1	0.368 ± 0.024	3.081 ± 0.040	3.908 ± 0.046	-0.040
Pooled	t_2	0.408 ± 0.032	2.950 ± 0.080	3.783 ± 0.085	
Pooled	t_2 + earlier lit. block	0.407 ± 0.031	2.958 ± 0.080	3.787 ± 0.083	-0.001
p-i-n	t_1	0.244 ± 0.058	3.168 ± 0.049	4.085 ± 0.114	-0.047
p-i-n	t_2	0.291 ± 0.065	3.047 ± 0.045	3.955 ± 0.131	
p-i-n	t_2 + earlier lit. block	0.294 ± 0.074	3.051 ± 0.067	3.947 ± 0.142	+0.002
n-i-p	t_1	0.374 ± 0.016	3.062 ± 0.047	3.918 ± 0.061	-0.029
n-i-p	t_2	0.404 ± 0.013	2.966 ± 0.036	3.825 ± 0.060	
n-i-p	t_2 + earlier lit. block	0.405 ± 0.010	2.964 ± 0.017	3.821 ± 0.045	+0.001

Feature-group controls (same protocol):

Subset	Variant	Feat.	R^2	MAE	RMSE
Pooled	t_1 + earlier lit. block (no material properties)	121	0.372 ± 0.026	3.060 ± 0.053	3.895 ± 0.061
Pooled	t_2 + earlier block minus 2 completeness flags	219	0.407 ± 0.028	2.953 ± 0.075	3.787 ± 0.078
Pooled	t_2 + 32 noise columns	211	0.404 ± 0.029	2.970 ± 0.065	3.796 ± 0.076
p-i-n	t_1 + earlier lit. block (no material properties)	122	0.251 ± 0.071	3.150 ± 0.058	4.065 ± 0.136
p-i-n	t_2 + earlier block minus 2 completeness flags	220	0.292 ± 0.070	3.052 ± 0.061	3.951 ± 0.134
p-i-n	t_2 + 32 noise columns	211	0.288 ± 0.072	3.067 ± 0.063	3.962 ± 0.144
n-i-p	t_1 + earlier lit. block (no material properties)	120	0.378 ± 0.014	3.050 ± 0.038	3.907 ± 0.053
n-i-p	t_2 + earlier block minus 2 completeness flags	218	0.407 ± 0.012	2.960 ± 0.024	3.816 ± 0.051
n-i-p	t_2 + 32 noise columns	210	0.402 ± 0.009	2.976 ± 0.025	3.830 ± 0.053

Table S6. Earlier-block rows of the reference-model, model-development and stricter-protocol experiments

Reference models, pooled, five DOI-grouped splits (mean ± s.d.):

Model	Tier	R^2	MAE	RMSE
XGBoost (defaults)	t_2 + earlier lit. block	0.377 ± 0.025	3.011 ± 0.057	3.880 ± 0.056
Random forest	t_2 + earlier lit. block	0.352 ± 0.021	3.073 ± 0.049	3.958 ± 0.070
MLP	t_2 + earlier lit. block	0.376 ± 0.023	3.042 ± 0.052	3.883 ± 0.056
GPR (3,000-record subsample)	t_2 + earlier lit. block	0.352 ± 0.029	3.129 ± 0.051	3.957 ± 0.065

Model-development experiment (R^2, mean ± s.d. over five splits):

Subset	Tier	Tuned	Bag	XGB	Ens
Pooled	t_2 + earlier lit. block	0.408 ± 0.029	0.412 ± 0.029	0.411 ± 0.024	0.418 ± 0.026
p-i-n	t_2 + earlier lit. block	0.295 ± 0.071	0.299 ± 0.069	0.284 ± 0.077	0.300 ± 0.074
n-i-p	t_2 + earlier lit. block	0.407 ± 0.015	0.409 ± 0.015	0.403 ± 0.015	0.413 ± 0.015

Stricter protocols, pooled CatBoost R^2 (mean ± s.d.):

Protocol	Tier	R^2	MAE
Lead-author-grouped	t_2 + earlier lit. block	0.395 ± 0.013	2.92
Forward in time	t_2 + earlier lit. block	0.335 ± 0.058	3.21

Figs. S1 and S2. SHAP analysis (models trained on t_2 + the earlier 44-predictor literature block; not t_3)

Table S7. Common-split comparison: per-seed paired ΔR^2

Reserved publications, matched default CatBoost; each seed model scored separately (the main-text values use the mean of the five seed models).

Population	Comparison	seed 0	seed 1	seed 2	seed 3	seed 4
LLM-covered	$t_2 - t_1$	+0.0427	+0.0431	+0.0427	+0.0414	+0.0415
LLM-covered	$t_3 - t_2$	-0.0012	+0.0120	+0.0059	+0.0045	+0.0092
All reserved	$t_2 - t_1$	+0.0323	+0.0337	+0.0328	+0.0356	+0.0301
All reserved	$t_3 - t_2$	+0.0012	+0.0079	+0.0017	-0.0023	+0.0034

Negative controls (default settings; ΔR^2 relative to t_2 under the same settings): LLM-covered: five DOI-permuted flag blocks -0.0049 to +0.0015; coverage indicator only +0.0015. all reserved: five DOI-permuted flag blocks -0.0028 to +0.0007; coverage indicator only +0.0011.

Extraction system prompt (verbatim from the recovered script)

```
You are a materials-science data-extraction assistant.
```

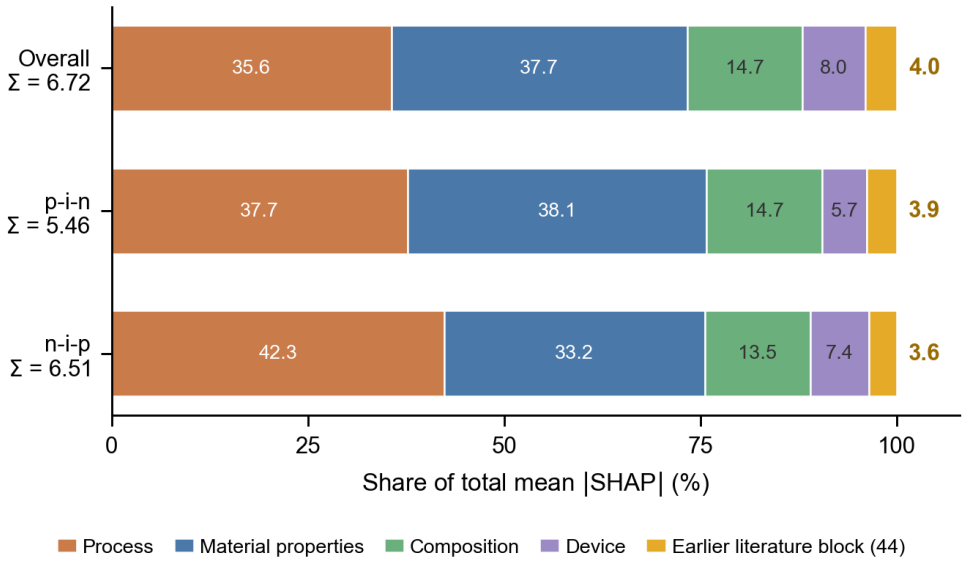


Figure 1: *

Fig. S1. Share of total feature attribution by feature group for matched CatBoost models trained on t_2 plus the earlier 44-predictor literature block (first DOI-grouped split, 1,500 test records). Attribution is not predictive benefit, and no SHAP analysis was run for the recipe-flag model t_3.

```
Given a Markdown document converted from a published paper, extract every
material reported with its measured or computed properties.

Return ONLY a single JSON object (no markdown fences, no commentary) with this
top-level schema:

{
  "source": "DOI: <doi-from-paper>",
  "materials": [
    {
      "formula": "<chemical formula>",
      "common_name": "<human-readable name>",
      "type": "<e.g. perovskite, ETL, HTL, metal contact>",
      ... <all properties you can find: band_gap, PV_performance, deposition_method, etc.>
    }
  ]
}

Rules:
- Always include units and measurement method when available.
- Include error bars / standard deviations if reported.
- For PV performance, separate champion vs average and HTM-free vs with-HTM.
- If the DOI is embedded in the document, use it for the "source" field.
- Do NOT invent data. Only extract what is explicitly stated.
- Return raw JSON only - no surrounding text, no code fences.
```

Model-development grid

CatBoost depth/L2 pairs: (6,3), (8,3), (10,3), (8,10), (10,10), (8,1); learning rate 0.05; up to 3,000 iterations; early stopping 150 rounds on the validation set; three seeds (42, 7, 123) for the selected configuration; XGBoost: depth 8, learning rate 0.05, 1,500 trees, subsample 0.8, colsample 0.7, $\lambda = 3$, early stopping 100 rounds.

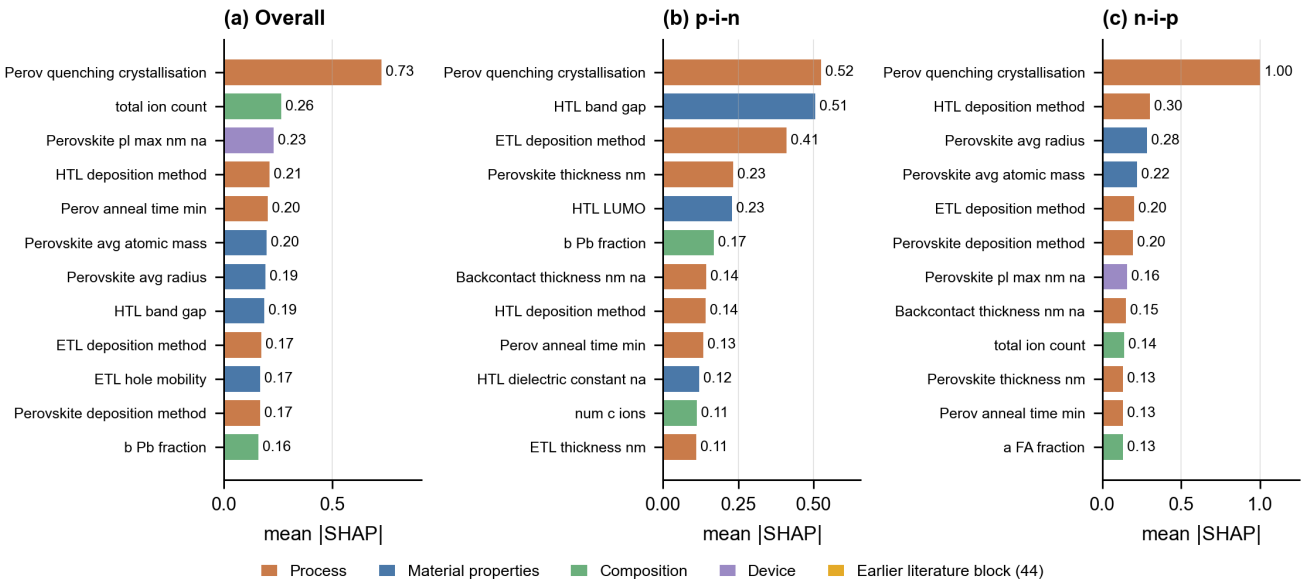


Figure 2: *

Fig. S2. Twelve highest-ranked features by mean |SHAP| (PCE percentage points) for the (a) pooled, (b) p-i-n and (c) n-i-p models of Fig. S1; `na` denotes a missingness indicator.